Lab-6: To solve the initial value problem $\int_0^2 \frac{e^x * \sin x}{1+x^2} dx$ using: C programming
i) trapezoidal rule, ii) Simpson's 1/3 Rule,
iii) Simpsons 3/8 Rule

Objective: To understand and implement numerical integration techniques such as Trapezoidal Rule, Simpsons 1/3 Rule and simpsons to approximate definite integrals and implement the solution in C-Programming.

Theory:

i) Trapezoidal Rule

↳ The Trapezoidal Rule approximates the area under a curve by dividing the interval into small trapezoids. for n equal sub-intervals, the formula is:

$$I = \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1})].$$

ii) Simpson's 1/3 Rule

↳ Simpson's 1/3 Rule uses quadratic polynomials to approximate the function; It requires the number of sub-intervals (n) to be even. The formula is.

$$I = \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \cdots + y_{n-2}))]$$

iii) Simpson's 3/8 Rule

↳ Simpson's 3/8 Rule approximates the integral using cubic polynomials.
The intervals is divided into sub-intervals that are multiple of 3.
The formula is:

$$I = \frac{3h}{8}[(y_0 + y_n) + 3(y_1 + y_2 + \cdots y_{n-1}) + 2(y_3 + y_6 + y_9 + \cdots + y_{n-3})].$$

Algorithms:

↳ Trapezoidal Rule:

Step i) Input the function $f(x)$, intervals bounds a and b, and the number of sub-intervals n.

Step ii) Calculate $h = \frac{b-a}{n}$.

Step iii) Calculate $y_0 = f(a)$ and $y_n = f(b)$.

Step iv) Compute the sum of y_i for $i = 1$ to $n-1$.

Step v) Apply the formula:

↳ $I = \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1})]$.

Step vi) Output the result.

↳ Simpson's 1/3 Rule:

Step i): Input the function $f(x)$, intervals bounds a and b, and the number of sub-intervals n (ensure n is even).

Step ii) Calculate $h = \frac{b-a}{n}$.

Step iii) Compute $y_0 = f(a)$ and $y_n = f(b)$.

Step iv) Compute the sum of y_i for odd and even indices separately.

Step v) Apply the formula:

$$I = \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \cdots + y_{n-2})]$$

Step vi) Output the result.

↳ Simpson's $\frac{3}{8}$ Rule

Step i): Input the function $f(x)$, interval bounds a and b, and the number of subintervals n (ensure n is a multiple of 3).

Step ii): Calculate $h = \frac{b-a}{n}$.

iii) Compute $y_0 = f(a)$ and $y_n = f(b)$.

Stepsiv): Compute the sums of y_i for indices that are not multiple of 3 and indices that are multiples of 3.

Stepv) Apply the formula:

$$I = \frac{3h}{8}[(y_0 + y_n) + 3(y_1 + y_2 + \cdots y_{n-1}) + 2(y_3 + y_6 + y_9 + \cdots + y_{n-3}).$$

Stepvi): output the Result.

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